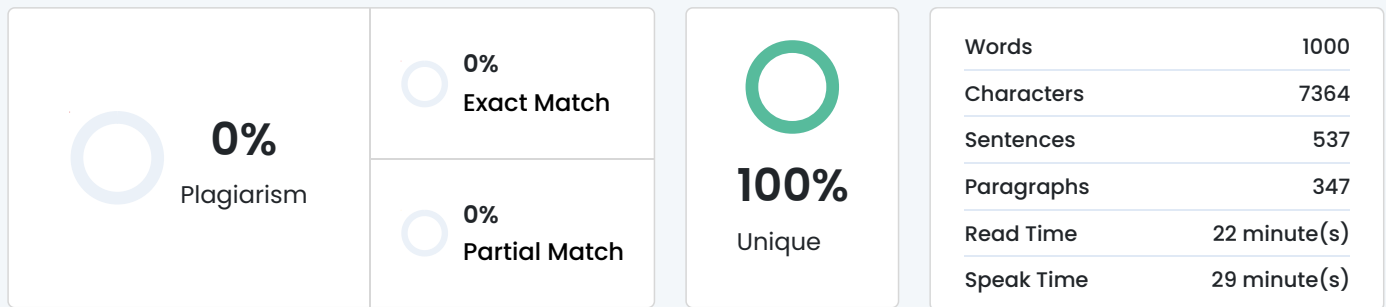


Plagiarism Scan Report



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GlucoSense: A Machine Learning-Based Web Application for Diabetes Risk Prediction and Explainable Health Assessment

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Abstract— Type-2 diabetes risk assessment can provide a better understanding of an individual's risk than a simple yes-or-no prediction. This paper presents GlucoSense, a web-based system developed as part of a reverse-engineering project for diabetes risk assessment and personalized health recommendations. The system uses the PIMA Indians Diabetes Database, which contains 768 records, with seven machine-learning features selected from the dataset. It supports three classification models: Logistic Regression, Random Forest, and XGBoost. The models are used independently, without combining their predictions through ensembling or stacking. During preprocessing, zero values in selected attributes are treated as missing values. Median imputation and five-neighbor KNN imputation are then applied, followed by standardization of the seven features using a preprocessor fitted on the training data. In addition to the machine-learning prediction, GlucoSense provides four rule-based scores: Clinical, Lifestyle, Family History, and Behavioral Risk

Scores. These scores are calculated separately and do not influence the machine-learning prediction. SHAP-based post-hoc explanations are also provided for XGBoost to help interpret its predictions. In the evaluation, Random Forest achieved an accuracy of 0.7922, while XGBoost obtained a test ROC-AUC of 0.8350 and the highest recall of 0.7593. The application consists of a web frontend, Java Spring Boot backend, Python FastAPI machine-learning service, file-based H2 storage, historical trend analysis, recommendations, and client-side PDF report generation. GlucoSense is intended for diabetes risk assessment and early awareness and should not be used as a clinical diagnostic tool.

Keywords— Type-2 Diabetes, Machine Learning, Risk Prediction, XGBoost, Random Forest, Logistic Regression, Explainable AI, SHAP, PIMA Indians Diabetes Database, Health Risk Assessment, Web Application.

I. Introduction

Diabetes is a health condition associated with high blood glucose levels and can cause complications over time. According to the World Health Organization (WHO), type-2 diabetes is the most common form of diabetes. WHO also emphasizes the importance of prevention, screening, treatment, and proper management of diabetes [1]. In this context, computer-based methods can support diabetes risk assessment by identifying patterns in health-related data before a person receives a medical diagnosis. However, such systems are intended to support risk identification and awareness, rather than replace clinical diagnosis or medical evaluation.

Diabetes is a health condition related to blood glucose levels and can cause different health complications over time. According to the World Health Organization (WHO), type-2 diabetes is the most common form of diabetes. The organization also highlights the importance of prevention, screening, treatment, and proper management of the condition [1]. In this context, computer-based methods can be used to assess diabetes risk by identifying patterns in health-related data before a person receives a medical diagnosis. However, these systems are intended only to support risk identification and awareness and should not be considered a replacement for clinical diagnosis.

GlucoSense is designed to meet the functional requirements of a web-based diabetes risk assessment system. It provides machine-learning-based risk prediction along with post-hoc explanations where available. The system also uses four rule-based risk dimensions to assess different aspects of risk, identifies relevant risk factors, and provides personalized recommendations. Users can view their previous predictions and analyse changes in risk over time. A web-based interface is provided for user interaction with the system. The project mainly focuses on implementation and reverse engineering. GlucoSense has not been clinically validated and is not intended to diagnose any medical condition. It should not be used as a replacement for professional medical advice or clinical diagnosis.

II. Related Work

GlucoSense is developed to meet the functional requirements of a web-based diabetes risk assessment system. It provides machine-learning-based risk prediction along with post-hoc explanations where available. The system also uses four rule-based risk dimensions to identify risk factors and provide personalized recommendations. Users can view their previous prediction results and analyse risk trends over time. The system provides a web-based interface for interacting with these features. The project mainly focuses on implementation and reverse engineering. GlucoSense has not been clinically validated and is not intended to diagnose any medical condition. It should not be used as a substitute for professional medical advice or diagnosis.

Scikit-learn provides commonly used preprocessing tools and traditional machine-learning algorithms [6]. Recent studies have also focused on making diabetes prediction models easier to interpret. Kibria et al. explored an explainable diabetes prediction workflow based on voting [8]. Khan et al. compared different diabetes risk prediction models with a focus on interpretability in a 2026 Scientific Reports article [9]. Netayawijit et al. developed a diabetes prediction framework that uses SHAP explanations to improve the understanding of model predictions [10]. These studies indicate an increasing focus on combining prediction performance with model interpretability. However, the datasets, preprocessing methods, model combinations, and evaluation approaches used in these studies differ from those implemented in GlucoSense.

Existing systems may have one or more limitations, such as relying on a single model, providing limited explanations for individual predictions, or not offering personalized recommendations. Some systems also do

not maintain prediction history for analysing changes over time or connect machine-learning results with broader lifestyle-related information. These limitations were considered while designing GlucoSense. However, GlucoSense does not claim to overcome all the limitations of previous systems. It currently uses a benchmark dataset and provides reliable SHAP explanations only for XGBoost models. In addition, the system has not yet been clinically validated.

III. Problem Statement

Many diabetes prediction systems mainly focus on generating a prediction from a limited number of clinical variables. However, the prediction alone may not give users enough information about the factors that contribute to the result. Some prediction interfaces also do not store previous assessments, making it difficult to

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